

Ayush Yadav

+91 9236593446 ayushthe8055@gmail.com LinkedIn GitHub

EDUCATION

Indian Institute of Technology, Patna

Bachelor of Technology in Computer Science and Engineering

August 2024 – May 2028

CPI: 8.71

Lucknow Public School, Lucknow, UP

Central Board of Secondary Education

April 2012 – May 2023

Percentage: 96.6%

COMPETITIVE PROGRAMMING

- **Specialist** on **Codeforces** with peak rating **1470** [Profile Link](#)
- **Knight** on **LeetCode** with peak rating **1886** [Profile Link](#)
- **3 Star Coder** on **Codechef** with peak rating **1624** [Profile Link](#)
- Solved over **1400+** algorithmic problems across **Codeforces**, **LeetCode**, **CodeChef**, and **AtCoder**.

ACHIEVEMENTS

- **ICPC Prelims 2025:** Secured **Rank 384**, solved **4 problems**, Team: **XNOR** (Team ID: **icpcpr251399**)
- Secured an **All India Rank 3941** out of **1.4 million+** candidates in **JEE Main 2024**
- Secured an **All India Rank 5581** out of **180,000** candidates in **JEE Advanced 2024**

PROJECTS

FixTown — Civic Issue Reporting Platform | Source Code

[Live Demo](#)

Tech Stack: React, Node.js, Express, PostgreSQL, PostGIS, Leaflet.js, Cloudinary, JWT, Vercel, Supabase

- Built a full-stack civic-tech platform enabling citizens to geo-tag and report infrastructure issues (potholes, manholes, utilities) with photo uploads, and municipal officers to track, prioritize, and resolve them.
- Implemented PostGIS geospatial queries for location-based issue filtering, JWT role-based authentication (citizen/officer) with invite-code validation, and Cloudinary CDN for auto-optimized image delivery.
- Developed community upvoting with optimistic UI, real-time in-app notifications, interactive Leaflet maps with custom markers, and officer analytics dashboards with resolution time tracking and category breakdowns.
- Deployed on Vercel (frontend) and Render (backend) with Supabase PostgreSQL, achieving a production-ready architecture with role-protected routes, bcrypt password hashing, rate limiting, and secure JWT token management.

Two-Pass Assembler and Emulator | Source Code

Tech Stack: C, GCC

- Developed a two-pass assembler that translates a custom assembly language into 32-bit machine code while generating object, listing, and log files.
- Implemented symbol table management, label resolution, assembler directives (**data**, **SET**), and comprehensive syntax and semantic error detection.
- Built a machine emulator supporting instruction execution, trace generation, memory dumps, and runtime error handling including illegal opcodes and infinite loop detection.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS, SQL

Frameworks/Technologies: React.js, Node.js, Express.js, MongoDB

Tools/Platforms: Git, GitHub, MongoDB Atlas, L^AT_EX

RELEVANT COURSEWORK

Data Structures and Algorithms, Computer Architecture, Database Management Systems, Discrete Mathematics, Linear Algebra, Statistics and Probability